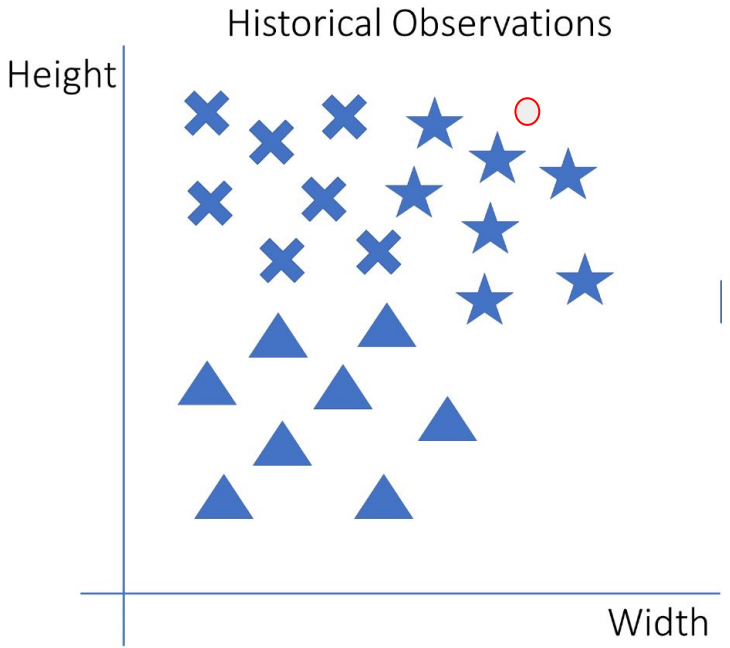
Recitation 3

Naive Bayes, KNN

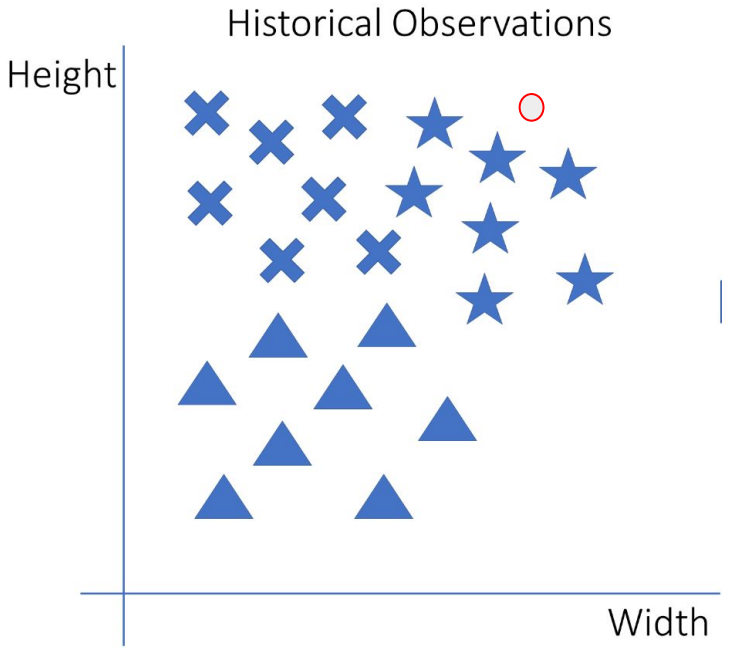
kNN (k-nearest neighbors)

1. Given point labels, and a new point, calculate the distance to all points.
2. k-nearest points determine the classification.
3. What is the classification of the red point using k=3?



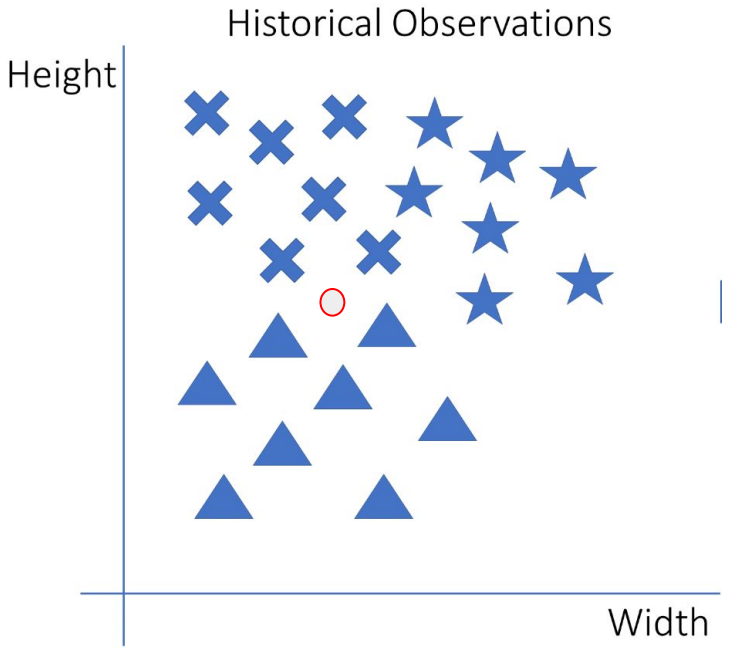
kNN (k-nearest neighbors)

1. Given point labels, and a new point, calculate the distance to all points.
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4. **Star.**



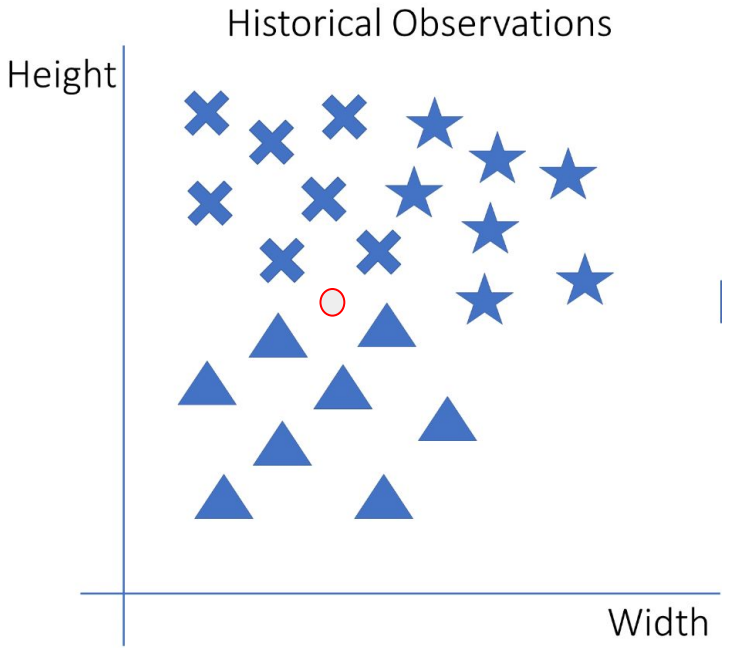
kNN (k-nearest neighbors)

1. Given point labels, and a new point, calculate the distance to all points.
2. k-nearest points determine the classification.
3. What is the classification of the red point using k=6?



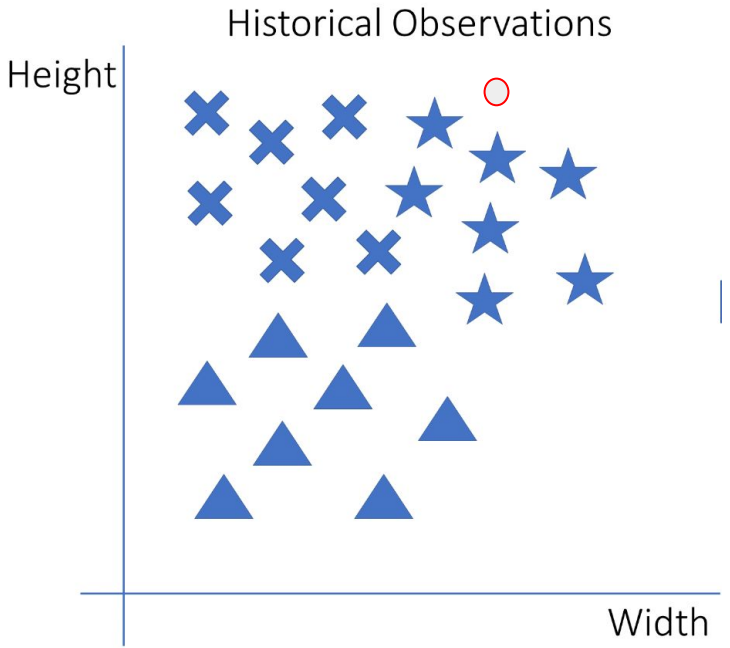
kNN (k-nearest neighbors)

1. Given point labels, and a new point, calculate the distance to all points.
2. k-nearest points determine the classification.
3. What is the classification of the red point using k=6?
4. **Ambiguous. Tiebreaker needed.**



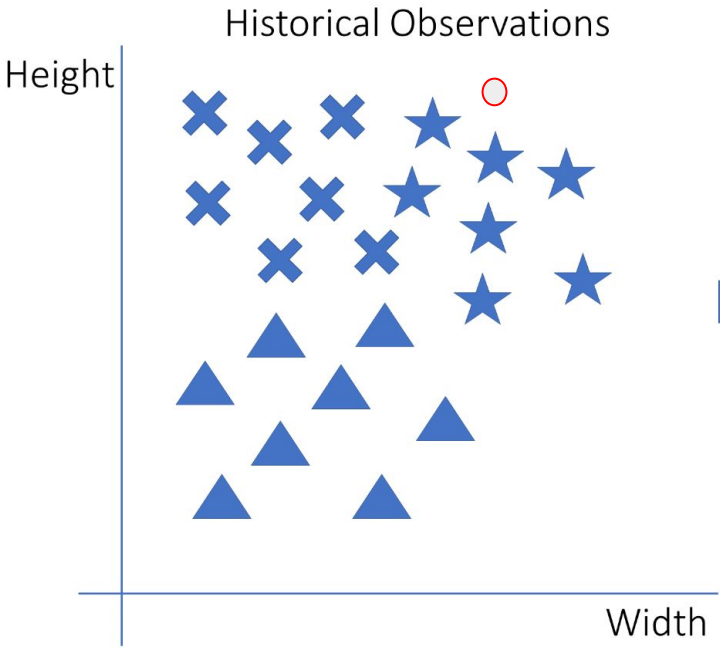
kNN (k-nearest neighbors)

1. Given point labels, and a new point, calculate the distance to all points.
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3. What is the classification of the red point using k=21?



kNN (k-nearest neighbors)

1. Given point labels, and a new point, calculate the distance to all points.
2. k-nearest points determine the classification.
3. What is the classification of the red point using k=21?
4. **Triangle. There's 21 points total.**



Calculating kNN

1. How to calculate kNN efficiently?
2. First we need to compute distance from point to every other point.
3. What’s the time complexity of that? (Hint: we have N points and p features per point).

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4. **O(p*N)**
5. Next, find the k-smallest values. How to do efficiently?
6. **Sort them.** O(N*Log(N))

Naïve Bayes classifier

Assumption: Conditional Independence

If X_1 and X_2 are conditionally independent given Y

$$P(X_1, X_2|Y) = P(X_1|X_2, Y)P(X_2|Y)$$

Chain rule

$$= P(X_1|Y)P(X_2|Y)$$

Conditional Independence

Naïve Bayes classifier

Benefits of Conditional Independence?

Genotype data X and healthy/cancer outcome Y

Individual	Locus 1 X_1	Locus 2 X_2	Locus 3 X_3	Healthy/Cancer Y
1	0	0	1	1
2	1	0	2	1
3	0	2	0	1
4	2	0	0	0
5	2	1	2	0
6	1	2	1	0

$$P(X_1 \dots X_n|Y) = \prod_i P(X_i|Y)$$

Naïve Bayes classifier

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$$P(X_1 \dots X_n|Y) = \prod_i P(X_i|Y)$$

(3^n – 1) x 2 parameters

2 x n x 2 parameters

Naïve Bayes classifier

What if X_i is continuous

Common approach: assume $P(X_i \mid Y=y_k)$ follows a Normal (Gaussian) distribution

- Assume each category y_k has a different mean and variance
- Perform MLE/MAP estimation

Naïve Bayes classifier: Example

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

naive Bayes Classifier:

P(Refund=Yes|No) = 3/7
P(Refund=No|No) = 4/7
P(Refund=Yes|Yes) = 0
P(Refund=No|Yes) = 1
P(Marital Status=Single|No) = 2/7
P(Marital Status=Divorced|No)=1/7
P(Marital Status=Married|No) = 4/7
P(Marital Status=Single|Yes) = 2/7
P(Marital Status=Divorced|Yes)=1/7
P(Marital Status=Married|Yes) = 0

For taxable income:
If class=No: sample mean=110
sample variance=2975
If class=Yes: sample mean=90
sample variance=25

Naïve Bayes classifier: Example

Given a Test Record: $X = (\text{Refund} = \text{No}, \text{Married}, \text{Income} = 120\text{K})$

- P(X|Class=No) = P(Refund=No|Class=No)
  × P(Married| Class=No)
  × P(Income=120K| Class=No)
  = 4/7 × 4/7 × 0.0072 = 0.0024

$$P(Income = 120 \mid No) = \frac{1}{\sqrt{2\pi}(54.54)} e^{\frac{(120-110)^2}{2(2975)}} = 0.0072$$

- P(X|Class=Yes) = P(Refund=No| Class=Yes)
  × P(Married| Class=Yes)
  × P(Income=120K|
  Class=Yes)
  = 1 × 0 × 1.2 × 10^{-9} = 0

Since P(X|No)P(No) > P(X|Yes)P(Yes)

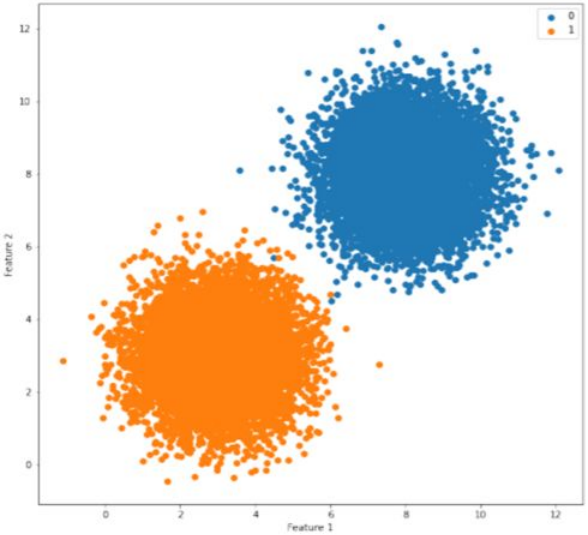
Therefore P(No|X) > P(Yes|X)

=> Class = No

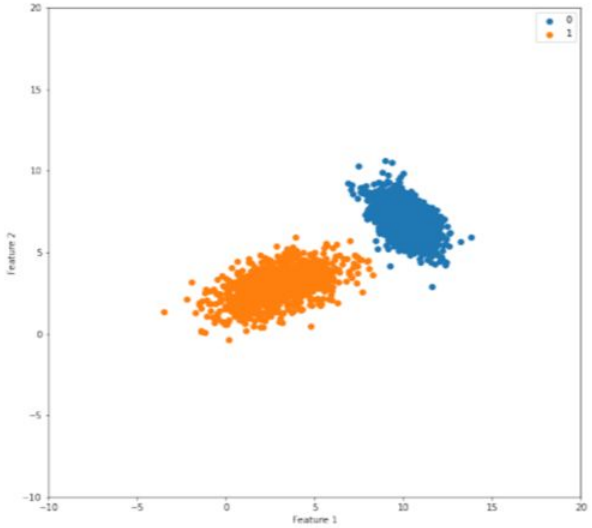
Is Naïve Bayes Assumption Valid?

Gaussian Contour Plots

Distribution: Multivariate (X1, X2) Gaussian



Gaussian Contour Plot of Dataset 1



Gaussian Contour Plot of Dataset 2